

INVESTIGATION REPORT

Issue 410: Debug request at reset must halt the core before it runs (closed bug, replayed)

openhwgroup/cv32e40x

Document	RTL-TRIAGE-CV32E40X-410
Disposition	Bug confirmed, fix verified
Record date	2026-09-11
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40x#410
Issue type	Bug report from an integrator or user
Affects	cv32e40x
Configuration	pre-fix RTL b2ef11e9 (2022-01-26) vs master; X_EXT=0
Disposition	Bug confirmed, fix verified
Severity	Not assessed
Agreement	Same disposition as the independent human record.
Evidence	6 cited, 4 present, 2 not retained

Disposition meaning. The reported behaviour was reproduced, and the checks listed under Evidence pass against the corrected design.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

Confirmed on the pre-fix RTL and verified fixed on master. Property: with debug_req_i held high from reset release (and fetch enabled), the debug state machine must leave HAVERESET directly into HALTED; RUNNING may only be visited after a HALTED has been seen (i.e. after a dret). On RTL b2ef11e9 (2022-01-26) BMC finds a 4-cycle counterexample: the controller passes through BOOT_SET into RUNNING at cycle 4 and only reaches HALTED at cycle 7 (cover). On master the property holds for all time (abc pdr, 8 s) and in a 25-cycle BMC; the cover shows HALTED reached at cycle 5 with no instruction retired. Three checker iterations: v1 assumed the controller-internal debug_req (a registered copy) and was unsatisfiable, v2 forbade RUNNING entirely (too strong: RUNNING is legal after a dret; master failed it at cycle 9 via a dret from the halt address), v3/v4 is the correct ordering property. Comparison with the maintainers' fix, read afterwards (PR #747, merged 2022-12-22): 'Removed BOOT_SET state from controller_fsm. Checking for debug_req_i in RESET state to enable going directly to debug mode from RESET without fetching any instructions.' This is exactly the transition the counterexample shows missing.

3 Conditions

Not recorded as separate fields for this run. The conditions and environment assumptions that bound this result are stated inside the Description, and the configuration is given in Status.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Checker used

checker_410.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	counterexample	formal/prefix_bmc4/job/engine_0/trace.vcd	not kept	pre-fix: RUNNING visited at cycle 4 before HALTED (cex_prefix_debugfsm.png)
E2	proof	formal/prefix_cover3/sby.log	present	pre-fix: cover RUNNING at 4, HALTED at 7 (the detour)
E3	proof	formal/master_prove4/sby.log	present	master: abc pdr unbounded PASS (8 s)
E4	proof	formal/master_bmc4/sby.log	present	master: BMC depth 25 PASS
E5	proof	formal/master_cover3/sby.log	present	master: HALTED reached at cycle 5 (cover_master_direct_halt.png); RUNNING only at 9 after a dret
E6	diff	cores/cv32e40x_hist (git show 32e84067)	not kept	PR #747: remove BOOT_SET, check debug_req_i in RESET. Read after proving.

Paths are relative to runs/e40x-410. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.
- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- diff. Source or configuration difference.

Artefact(s) E1, E6 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

7 Limitations and scope

- The fix is verified against the checks listed under Evidence only. This is not a regression sign-off and does not cover the wider test suite.
- This is an agent investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.
- Public demonstration runs on open cores are not client engagements and not production verification results.

8 Provenance and measurement

Run directory	runs/e40x-410
Title source	Title and configuration read back from reports/issues/cv32e40x_issue_410.pdf, the previously published report for this issue.
Inputs shown	title, body
Human record	PR #747 (2022-12-22): removed BOOT_SET state, debug_req_i checked in RESET so the core enters debug directly without fetching
Agent confidence	0.9. This is the investigating agent's own estimate. It is not a measurement and not a reliability figure.
Iterations	4

Run by Claude directly

Measured tool and step time

Step	Measured
read issue + pre-fix checkout + wrapper	5.0 min (300 s)
checker v1 (unsat) + diagnose	4.0 min (240 s)
checker v2/v3 runs + diagnose dret	7.0 min (420 s)
checker v4 pre-fix CEX	60.0 s
master BMC + pdr	42.0 s
compare fix PR #747	60.0 s
report	60.0 s
Total	19.7 min (1182 s)

Times are wall-clock for the step named. Some steps may have run concurrently, so they do not sum to an isolated performance figure, and they are not an engineering effort measure.

Note. wall clock from first job to verdict; phases estimated from job timestamps

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